MILITARY SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPE 8951

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, hydrogen charging
See figure 1 and note 17
Mounting position: Any
Weight: 8-ounces nominal

ABSOLUTE RATINGS:

Parameter:	Ef	epy	epx	Ebb	egy	egx	Ecc	ib	Ib
Unit:	Vac	kv	kv	Vdc	v	v	Vdc	a	Adc
Maximum:	6.8	25	25 (Note 1)	---	1,500 (Note 2)	400	---	1,000 (Note 15)	2.5
Minimum:	5.8	---	5% epy	1,500	450	---	-200	---	---
TEST CONDITIONS: (Note 8)	6.3	25	---	---	450	---	---	---	---

ABSOLUTE RATINGS:

Parameter:	Ip	tj	prr	dik/dt	Pb	tk	TA	Cooling	Eres
Unit:	Aac	μs	pps	a/μs	---	sec	°C	---	Vac
Maximum:	47.5	0.005	4,000	5,000	25×10^9 (Note 7)	---	85	--- (Note 7)	6.8
Minimum:	---	---	---	---	---	300	-55	---	5.8
TEST CONDITIONS: (Note 8)	---	---	---	---	---	300	Ambient	---	6.3

ABSOLUTE RATINGS: Charging triode circuit (see figure 4)

Parameter:	ib	Rate of change of cathode voltage
Unit:	a	kv/μs
Maximum:	65 (Note 15)	-10 (Note 16)
Minimum:	---	---
TEST CONDITIONS: (Note 8)	---	---

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
3251	Emission	14	ik = 1,500 a; tp = 5 μs ±10%; prr = 60 ±10%; tr = 0.5 μs (max)	0.65	II	egk	---	250	v
3241	Heater current (cathode)	-		0.65	II	If	---	18	Aac
3241	Heater current (reservoir)	-		0.65	II	Ires	---	8	Aac
3246	Operation (1)	3, 4, 5	Ef = Eres = 5.8 Vac; epy = 24 kv; Ib = 2.2 Adc; tk = 300 sec (max); t = 1 hour	0.65	II	egy	---	450	v
3246	Operation (1A)	10	Operation (1) except Ef = Eres = 6.3 Vac; t = 5 hours	0.65	II	egy	---	450	v
3267	Instantaneous starting	3, 6	epy = 18 kv (min); Ef = Eres = 6.8 Vac	0.65	II	---	---	---	---
3247	DC anode voltage for conduction	3	Ef = Eres = 5.8 Vac; tk = 300 sec (max); egy = 450 v (max)	0.65	II	Ebb	---	1,000	Vdc
	Quality conformance inspection, part 2								
3256	Anode delay time	-	Operation (1) except t = 120 seconds	---	---	tad	---	0.5	μs
3256	Anode delay time drift	9	Anode delay time	---	---	Δtad	---	0.15	μs
3261	Time jitter	12	Operation (1A); tk = 300 sec	---	---	tj	---	0.005	μs
	Quality conformance inspection, part 3								
---	Life test	3, 11	Group C: t = 500 hours	---	---	---	---	---	---
---	Life-test end points:	-							
3246	Operation (1) and 1A, except note 5	-		---	---	Ebb tj egy	--- --- ---	1,250 0.005 450	Vdc μs v
	Periodic-check tests								
1031	Sweep-frequency vibration	-	No voltage applied	---	---	---	---	---	---
1041	Shock	-	250 G; no voltage applied	---	---	---	---	---	---
---	Shock and sweep-frequency vibration end point:								
	DC anode voltage for conduction	-		---	---	Ebb	---	1,250	Vdc

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NOTES:

1. In pulsed operation, the peak inverse voltage exclusive of a spike of 0.05 μs maximum duration, shall not exceed 5 kv during the first 25 μs following the anode pulse.

2. The driver pulse, measured at the tube socket with the thyratron grid disconnected: tr = 0.35 μs maximum; grid pulse duration = 2 μs minimum. The impedance of the drive circuit shall be 50 to 200 ohms.

3. The circuit used for this test is shown on figure 3. The anode circuit constants shall be chosen for a resonant charging condition of 4,000 pps minimum. At epy = 24 kv and ib = 580a minimum, the rate of rise of the current pulse (dik/dt) shall be 5,000 a/μs minimum. The pulse width (tp) shall be 2.4 μs ±10 percent. The grid pulse measurements at the tube socket with the thyratron grid disconnected shall be as follows: tr = 0.35 μs minimum, tp = 2.0 μs maximum, Zg = 200 ohms minimum, Ecc = -70 V minimum.

4. The anode temperature and grid seal temperature shall not exceed 350°C at any time during this test.

5. Anode voltage shall be applied as specified in note 6. Full anode voltage (epy = 24 kv) shall be reached within 5 seconds after starting. The tube shall operate at the specified anode voltage for a total of 6 hours which may include two interruptions during the initial 30 minutes and one interruption during the last 5-1/2 hours.

6. This test shall be the first test performed after the holding period. The tube shall operate satisfactorily on push button starting within two attempts when the anode voltage (epy) is applied to the tube under test in such a manner as to rise from 0 to 18 kv minimum within 0.03 second. The filter in the rectifier shall be designed so that epy reaches at least 9 kv within 0.015 second. The interval between successive attempts to instantaneously start the tube shall be not less than 10 seconds nor more than 30 seconds.

7. There shall be no artificial anode or envelope cooling directed onto the tube. Blowers or anode radiators shall not be used.

8. The heat dissipator, Hughes Aircraft Co. Part No. 529033 (grid radiator), or equivalent shall be used during all tests of this specification.

9. This test shall be performed simultaneously with operation (1) test. Anode delay time measurement shall be made at the end of 2, 4, and 30 minutes of the operation (1) tests. The anode delay time drift (Δtad) measurement is the numerical difference between 2 minutes and 4 minutes, or 2 minutes and 30 minutes anode delay time readings, whichever is greater.

10. Operation (1A) shall be a continuation of operation (1) with the specified change in operating conditions made after 1 hour.

11. Where production is less than 50 tubes per month, life-test sample size shall be 1 tube per month.

12. The tube shall be tested by applying a peak forward anode voltage, as specified in the test conditions for the time jitter test immediately after the tk. The tj shall be not greater than the amount specified after 120 seconds of operation.

13. The tube shall operate at the specified anode voltage for a total of 6 hours which may include three interruptions during the initial 30 minutes and three interruptions during the last 5-1/2 hours.

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NOTES: -Continued

14. The positive pulse shall be applied to the grid of the tube with the anode floating. Measure the voltage between grid and cathode not more than 2.5 μs after the beginning of the current pulse. The average voltage shall not increase from the voltage measurement point to the end of the pulse. As an alternate, the test may be conducted by connecting the grid to the anode through a 1-ohm resistance, applying the positive pulse and reading the peak voltage drop between anode and cathode (epk). The limit for this reading shall be the same as that for egk in the emission test.

15. The absolute maximum ib is 1,000a when used in the circuit of figure 3, with a pulse width (tp) of less than 70 μs. The absolute maximum ib is 65a when used in the circuit of figure 4, with a pulse width (tp) of greater than 400 μs.

16. When used in the circuit of figure 4, the absolute-maximum rate of change of cathode voltage at the time of switch tube trigger is -10 kv/μs.

17. An MT-4 mounting assembly (or equivalent) shall be provided as an integral part of each tube (see figure 2 for reference only).

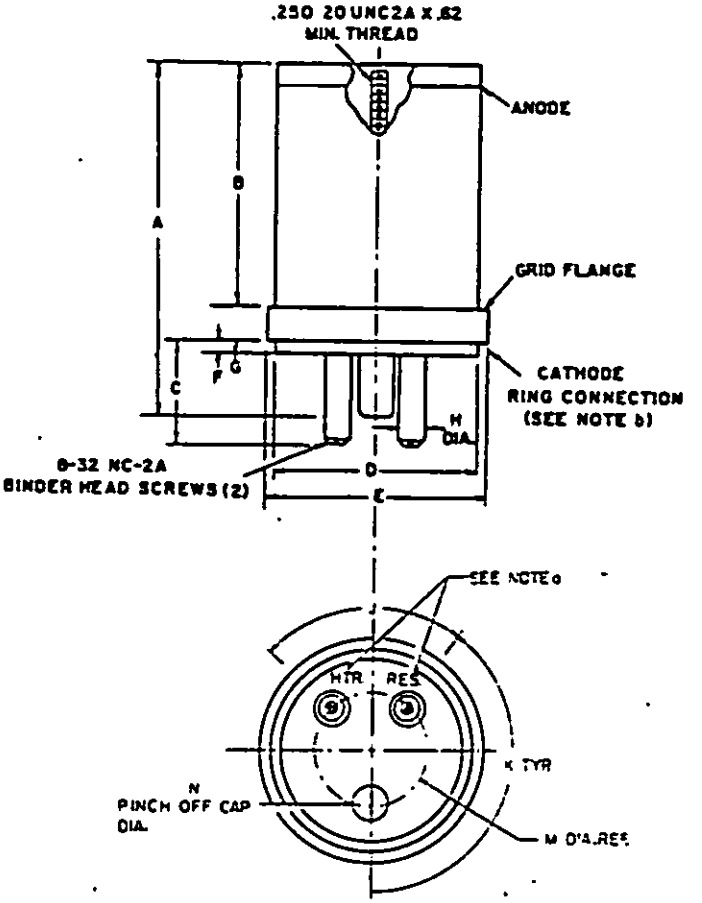
User activities:
Navy - AS, OS, MC, CG, SH

Preparing activity: Navy - EC

Agent: DSA - ES

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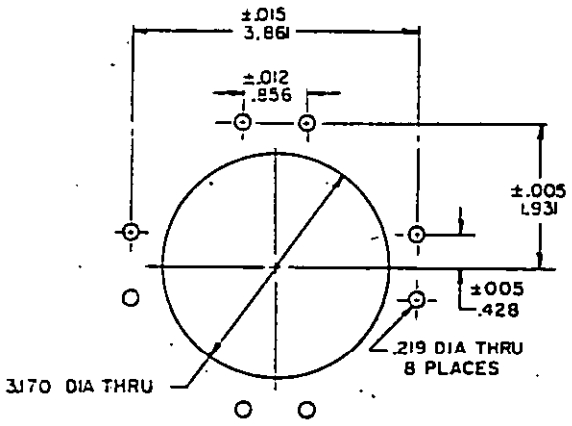


Ltr	Dimensions in inches with metric equivalents (mm) in parentheses.	
	Minimum	Maximum
Quality conformance inspection, part 2		
A		5.420 (137.67)
B		3.700 (93.98)
C		1.555 (39.50)
D	3.100 (78.74)	3.130 (79.50)
E	3.245 (82.42)	3.305 (83.95)
F	.120 (3.05)	.200 (5.08)
G	.520 (13.26)	.570 (14.48)
H		.375 (9.53)
N		.469 (11.91)
Reference dimensions		
J	90°	
K	135°	
M	1.50 (38.10)	

NOTES:
a. Reservoir and heater terminals shall be identified and marked on tube base as specified herein.
b. Other side of reservoir and heater terminals are internally connected and are common to cathode.

FIGURE 1. Outline drawing of electron tube type 8951.

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INCHES	MM
.005	.13
.012	.30
.015	.38
.190	4.83
.219	5.56
.428	10.87
.856	21.74
1.931	49.05
3.170	80.52
3.861	98.07

Mounting hole dimensions.

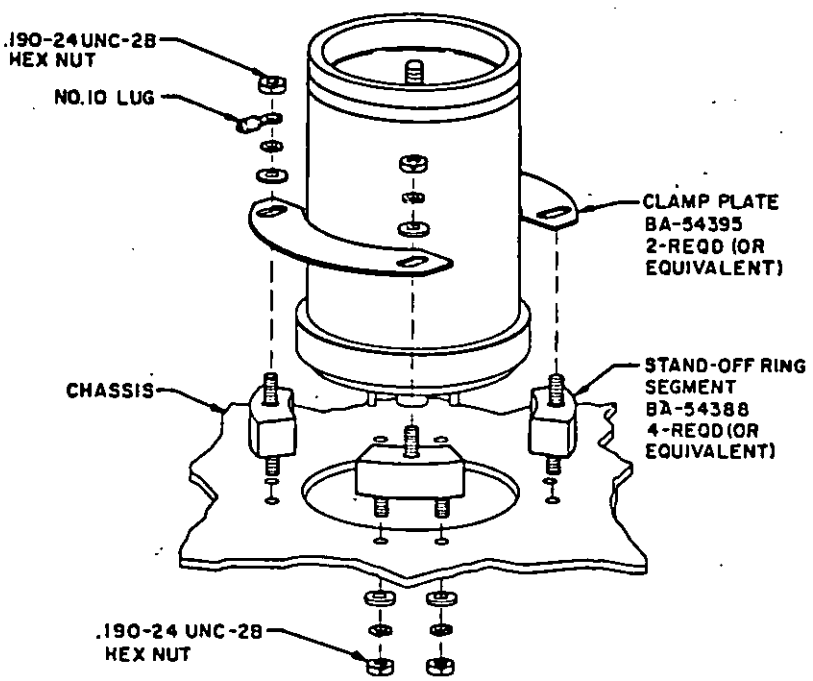


FIGURE 2. MT-4 tube mounting assembly.

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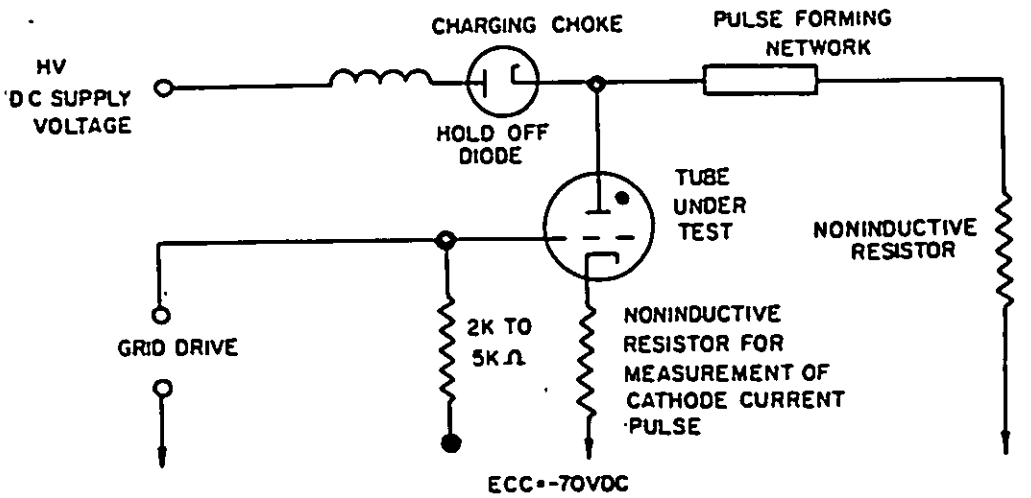
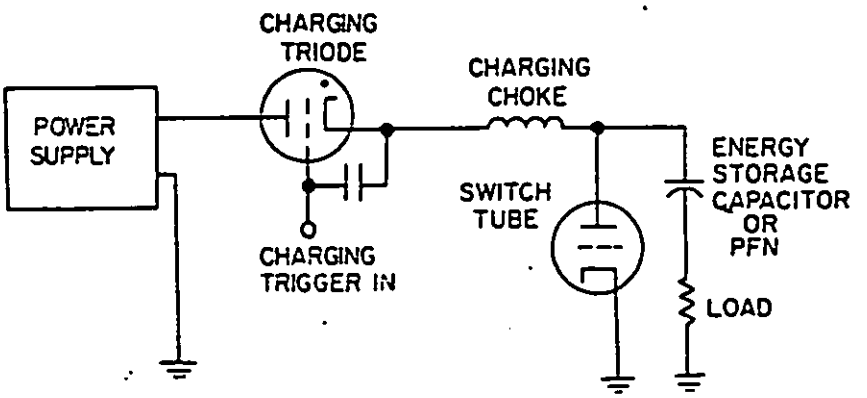


FIGURE 3. Basic soft tube modulator circuit.



TYPICAL CIRCUIT

FIGURE 4. Basic charging triode circuit.

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